Floor plan

10 x 10 m clear · buttress every 1.5 m · printed walls

1. Printed wall 400 (= roof base width), room side out: inner bead 30, lattice zone 310 (wavy bead 30 with insulation between its loops, dotted), 2 outer beads 30. Buttresses solid black: 400 wide (same as the wall), 1000 deep from the inner face (flush with the wall inside), so they project 600 outside; one about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included).
2. Every other bay (the 2nd, 4th and 6th) is a full-height curtain wall, the whole bay between two buttresses; the middle bay of the south and north walls has a 1000 door instead. The other bays stay printed wall. Positions and sizes are placeholders.
3. Pyramid roof above (dashed): 1000 x 1000 clear crown opening, hips to the corners; the roof is 400 thick square to the slope from eave to crown, so the shell ends 100 outside the wall face; the outer beads are exposed concrete (no cladding).
4. Buttress axes 1486 apart along each wall (centre to centre), corners included.

Notes

Section A-A cuts through the 4th buttress from the south on the east and west walls (y = -743).

Detail D1 is the eave at the east buttress on that cut.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP
Scale 1:50 at A3 · dimensions in mm
2026-10-14 · hand estimates, not engineered

01

Floor plan · roof grid A

10 x 10 m clear · buttress every 1.5 m · printed walls

1. Printed wall 400 (= roof base width), room side out: inner bead 30, lattice zone 310 (wavy bead 30 with insulation between its loops, dotted), 2 outer beads 30. Buttresses solid black: 400 wide (same as the wall), 1000 deep from the inner face (flush with the wall inside), so they project 600 outside; one about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included).
2. Every other bay (the 2nd, 4th and 6th) is a full-height curtain wall, the whole bay between two buttresses; the middle bay of the south and north walls has a 1000 door instead. The other bays stay printed wall. Positions and sizes are placeholders.
3. Pyramid roof above (dashed): 1000 x 1000 clear crown opening, hips to the corners; the roof is 400 thick square to the slope from eave to crown, so the shell ends 100 outside the wall face; the outer beads are exposed concrete (no cladding).
4. Buttress axes 1486 apart along each wall (centre to centre), corners included.
5. Cast ribs of roof grid A seen from above (dashed), with the bottom ring (the 300 solid base) and the horizontal / top rings.

Notes

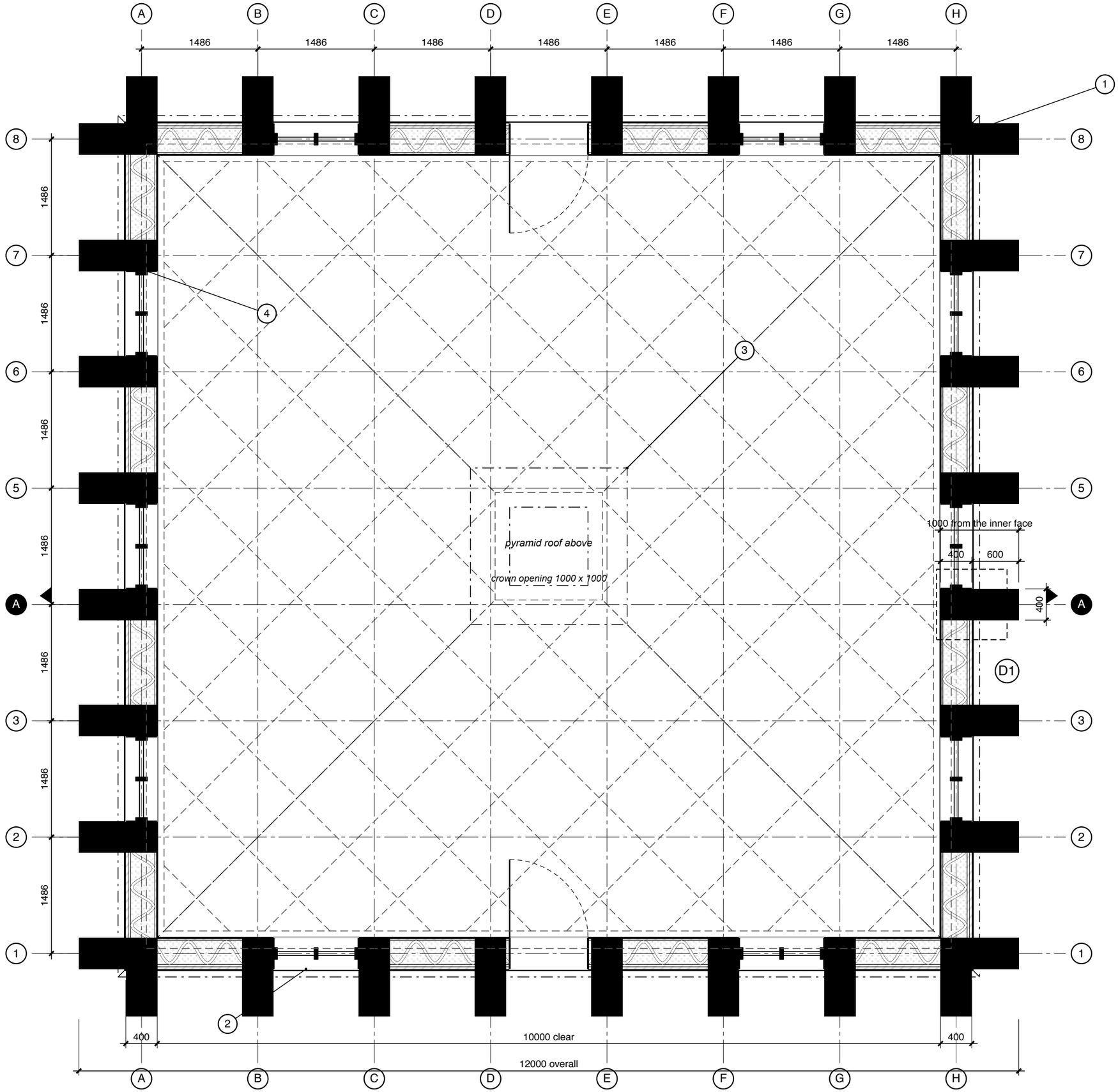
Section A-A cuts through the 4th buttress from the south on the east and west walls (y = -743).

Detail D1 is the eave at the east buttress on that cut.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP
Scale 1:50 at A3 · dimensions in mm
2026-10-14 · hand estimates, not engineered

02



Floor plan · roof grid B

10 x 10 m clear · buttress every 1.5 m · printed walls

1. Printed wall 400 (= roof base width), room side out: inner bead 30, lattice zone 310 (wavy bead 30 with insulation between its loops, dotted), 2 outer beads 30. Buttresses solid black: 400 wide (same as the wall), 1000 deep from the inner face (flush with the wall inside), so they project 600 outside; one about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included).
2. Every other bay (the 2nd, 4th and 6th) is a full-height curtain wall, the whole bay between two buttresses; the middle bay of the south and north walls has a 1000 door instead. The other bays stay printed wall. Positions and sizes are placeholders.
3. Pyramid roof above (dashed): 1000 x 1000 clear crown opening, hips to the corners; the roof is 400 thick square to the slope from eave to crown, so the shell ends 100 outside the wall face; the outer beads are exposed concrete (no cladding).
4. Buttress axes 1486 apart along each wall (centre to centre), corners included.
5. Cast ribs of roof grid B seen from above (dashed), with the bottom ring (the 300 solid base) and the horizontal / top rings.

Notes

Section A-A cuts through the 4th buttress from the south on the east and west walls (y = -743).

Detail D1 is the eave at the east buttress on that cut.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

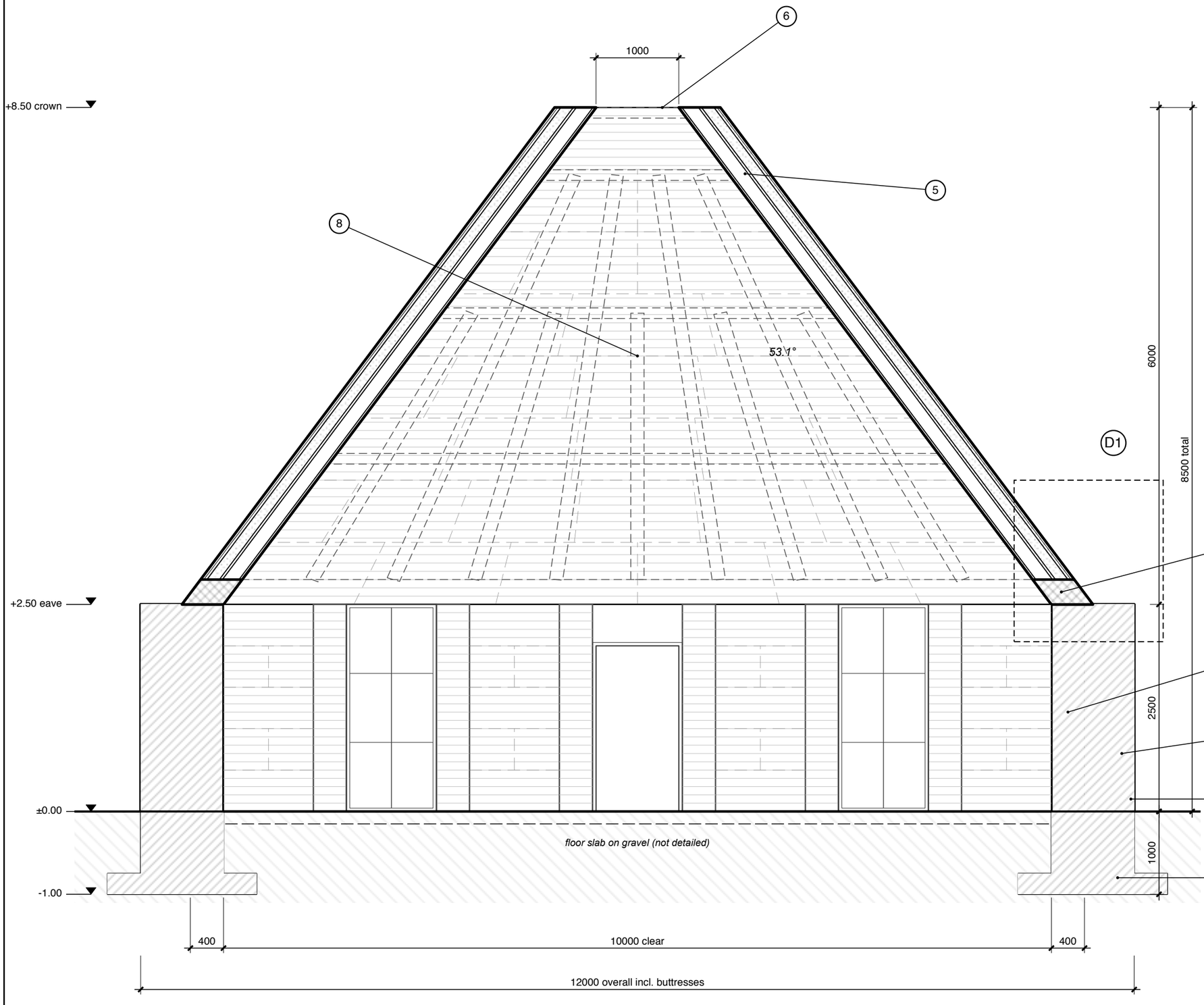
Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

03

Total section A-A · grid A

through the east and west buttresses, looking north



1. Footing 1800 x 250, underside at -1000 (frost depth and bearing are guesses).
2. Wall 400. The cut passes through a buttress, so the wall is solid with the buttress here (no insulation). Between buttresses it is printed: an inner bead 30, a lattice zone 310 (wavy bead + insulation), 2 outer beads 30.
3. Solid base of the roof: the first 300 (vertical) above the eave is completely solid concrete, 400 wide, no cavity or lattice zone; there is no ring beam.
4. Buttress 400 wide, 1000 from the inner face (600 beyond the wall), one about every 1.5 m (8 per side); it rises to +2.50, flush with the roof base.
5. Roof shell 400 thick square to the slope, the same from eave to crown (about 500 horizontally), every printed layer one 30 bead: inner bead 30, rib zone 160 (cavity crossed by the cast ribs), middle bead 30, lattice zone 120 (wavy bead + insulation, dotted), 2 outer beads 30, exposed concrete outside.
6. Crown opening 1000 clear; roof rises 6000 over 4500, slope 53.1° (36.9° from the vertical).
7. Everything is 3D printed in 10 mm layers; crossed hatch = solid cast concrete (roof base, ribs). Printed layer joints and the cast fill of the buttress and footing are shown in the roof-grid sheets and in detail D1.
8. Cast ribs of the roof grid, seen on the back face as dashed outlines (roof grid A, ruled-surface columns). The cut lies between ribs, where the rib zone is an empty cavity; rib sections are on the roof-grid sheets.

Notes

Roof build-up as in the Karamba model (scripts/karamba_model.py): bead 30, 2 outer beads, lattice 120, rib zone 160; inner and middle bead are the structural skins, the ribs the webs between them.

The cut runs through the 4th buttress from the south (y = -743), as on the floor plan. Beyond it the north wall is seen as on the plan: buttress every 1.5 m (strips outside the wall), every other bay printed wall, the bays in between full-height curtain wall (door in the middle one).

Faint grid on the back roof face and the north wall = printed pieces in running bond, about 1500 wide (assumed sizes). Dashed outlines on the back roof face = cast ribs.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

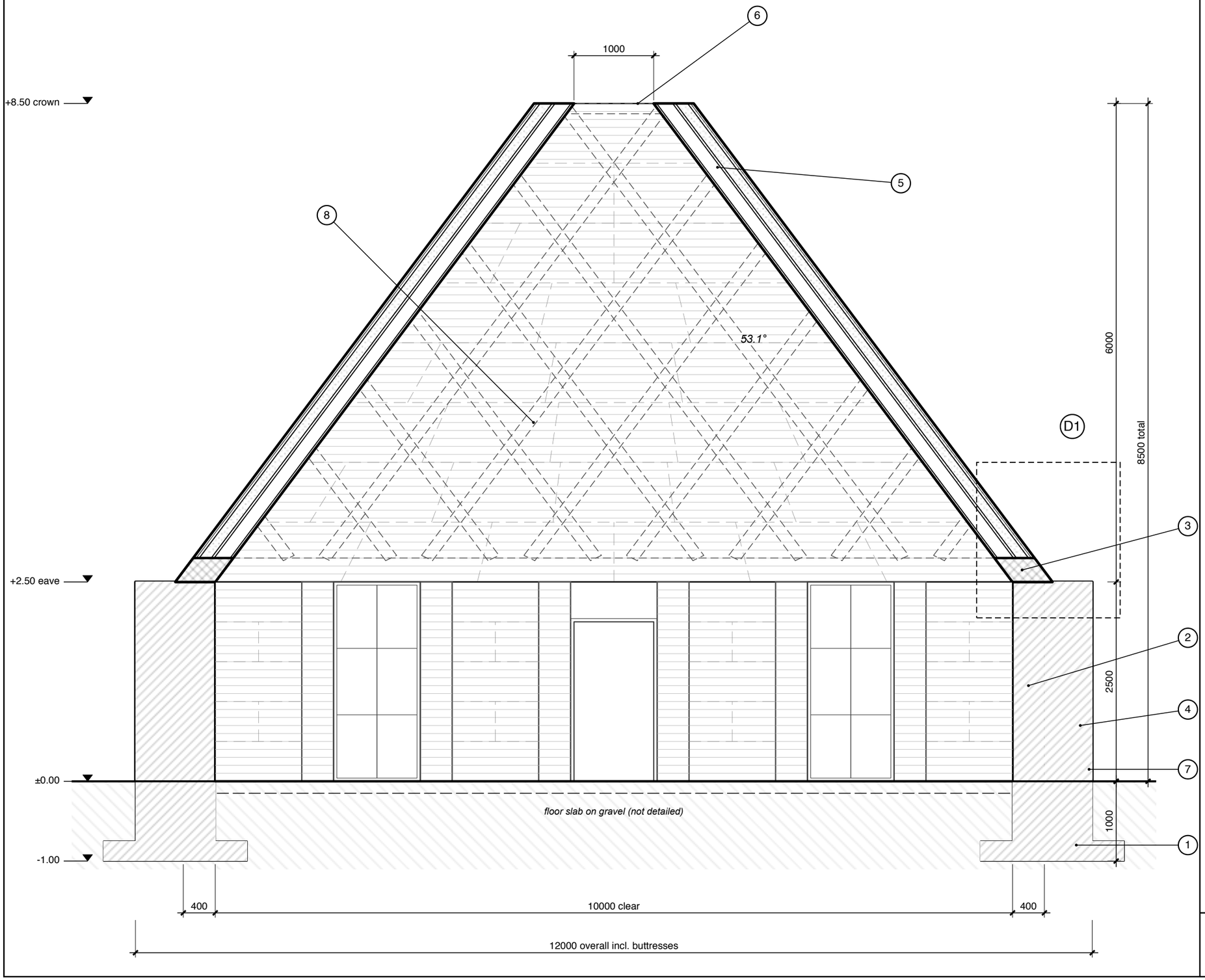
Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

04

Total section A-A · grid B

through the east and west buttresses, looking north



1. Footing 1800 x 250, underside at -1000 (frost depth and bearing are guesses).
2. Wall 400. The cut passes through a buttress, so the wall is solid with the buttress here (no insulation). Between buttresses it is printed: an inner bead 30, a lattice zone 310 (wavy bead + insulation), 2 outer beads 30.
3. Solid base of the roof: the first 300 (vertical) above the eave is completely solid concrete, 400 wide, no cavity or lattice zone; there is no ring beam.
4. Buttress 400 wide, 1000 from the inner face (600 beyond the wall), one about every 1.5 m (8 per side); it rises to +2.50, flush with the roof base.
5. Roof shell 400 thick square to the slope, the same from eave to crown (about 500 horizontally), every printed layer one 30 bead: inner bead 30, rib zone 160 (cavity crossed by the cast ribs), middle bead 30, lattice zone 120 (wavy bead + insulation, dotted), 2 outer beads 30, exposed concrete outside.
6. Crown opening 1000 clear; roof rises 6000 over 4500, slope 53.1° (36.9° from the vertical).
7. Everything is 3D printed in 10 mm layers; crossed hatch = solid cast concrete (roof base, ribs). Printed layer joints and the cast fill of the buttress and footing are shown in the roof-grid sheets and in detail D1.
8. Cast ribs of the roof grid, seen on the back face as dashed outlines (roof grid B, diamond lattice). The cut lies between ribs, where the rib zone is an empty cavity; rib sections are on the roof-grid sheets.

Notes

Roof build-up as in the Karamba model (scripts/karamba_model.py): bead 30, 2 outer beads, lattice 120, rib zone 160; inner and middle bead are the structural skins, the ribs the webs between them.

The cut runs through the 4th buttress from the south (y = -743), as on the floor plan. Beyond it the north wall is seen as on the plan: buttress every 1.5 m (strips outside the wall), every other bay printed wall, the bays in between full-height curtain wall (door in the middle one).

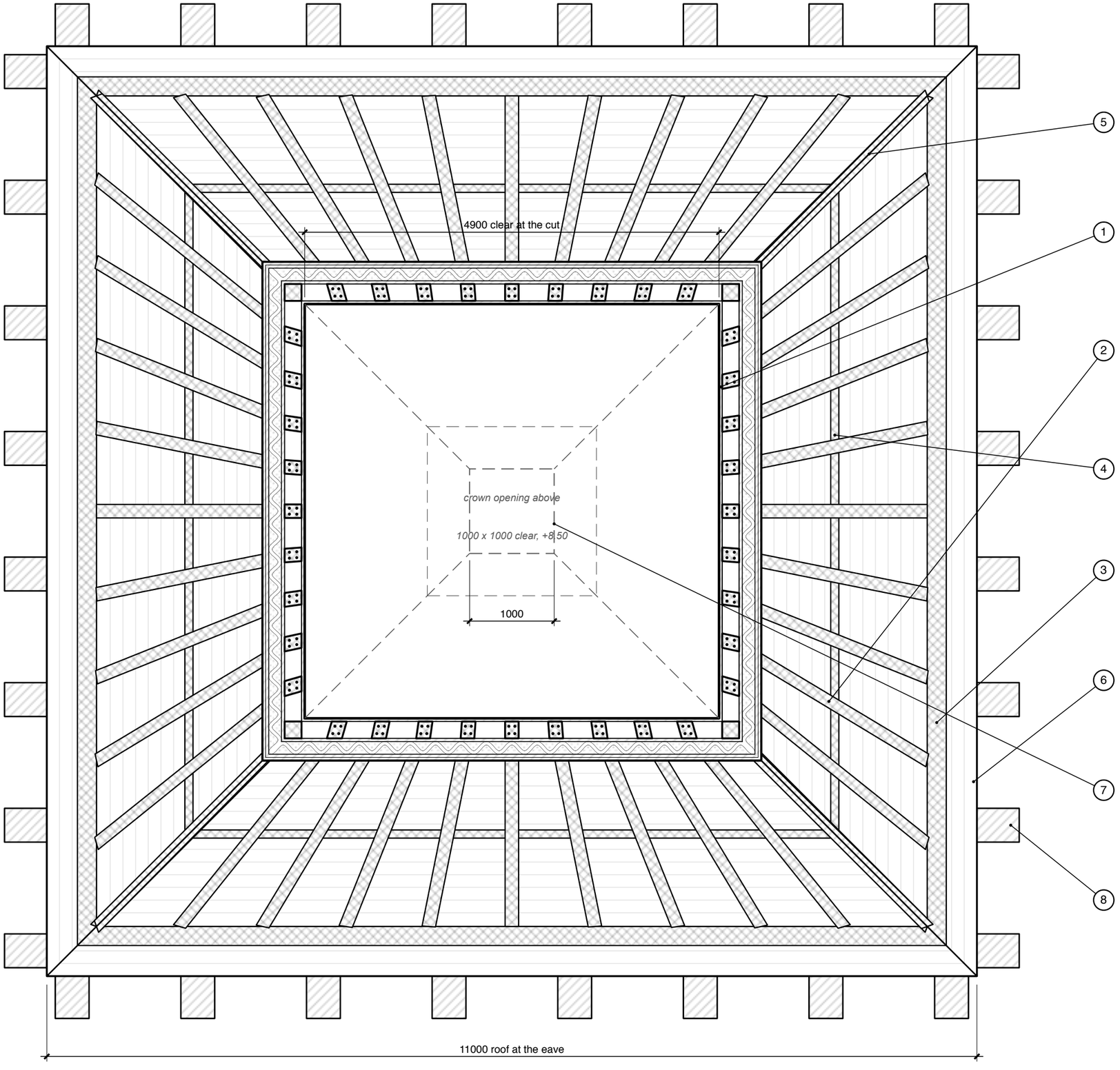
Faint grid on the back roof face and the north wall = printed pieces in running bond, about 1500 wide (assumed sizes). Dashed outlines on the back roof face = cast ribs.

0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered



Roof plan, cut at +5.90 · grid A

horizontal cut through the roof, looking down on the ribs

1. Cut at +5.90 (3400 above the eave, just below the horizontal rib at +6.02), room side out: inner bead 30, rib zone 160 with the cast ribs cut (crossed hatch; empty cavity between them), middle bead 30, lattice zone 120 (wavy bead + insulation, dotted), 2 outer beads 30 (exposed concrete).
2. Ribs of grid A seen below the cut: all 11 columns per face run from the base ring at the eave up to the cut (every other one stops at +6.02, just above it).
3. Base ring: the 300 solid base at the eave (+2.50 to +2.80).
4. Horizontal rib of grid A at +4.26 (the next ones, at +6.02 and +7.68, are above the cut).
5. Hips: ribs from the corners of the eave up to the cut; dashed above it, up to the crown opening.
6. Roof edge at the eave (outer face of the outer beads). Very faint squares = printed layer joints (real layers 10, drawn every 200 in height).
7. Above the cut (removed, dashed): hips and the crown opening at +8.50, 1000 clear inside (1275 at the rib axis, 2000 outside the outer beads).
8. Buttresses below the roof: their ends show beyond the roof edge; the wall under the roof is dashed.

Notes

Matches the roof-grid elevations: same rib geometry, same levels (+4.26, +6.02, +7.68 for the horizontal ribs of grid A).

The ribs lie in the rib zone of the shell, between the inner and the middle bead; here they are drawn as if the middle bead, the lattice zone and the outer beads were transparent.

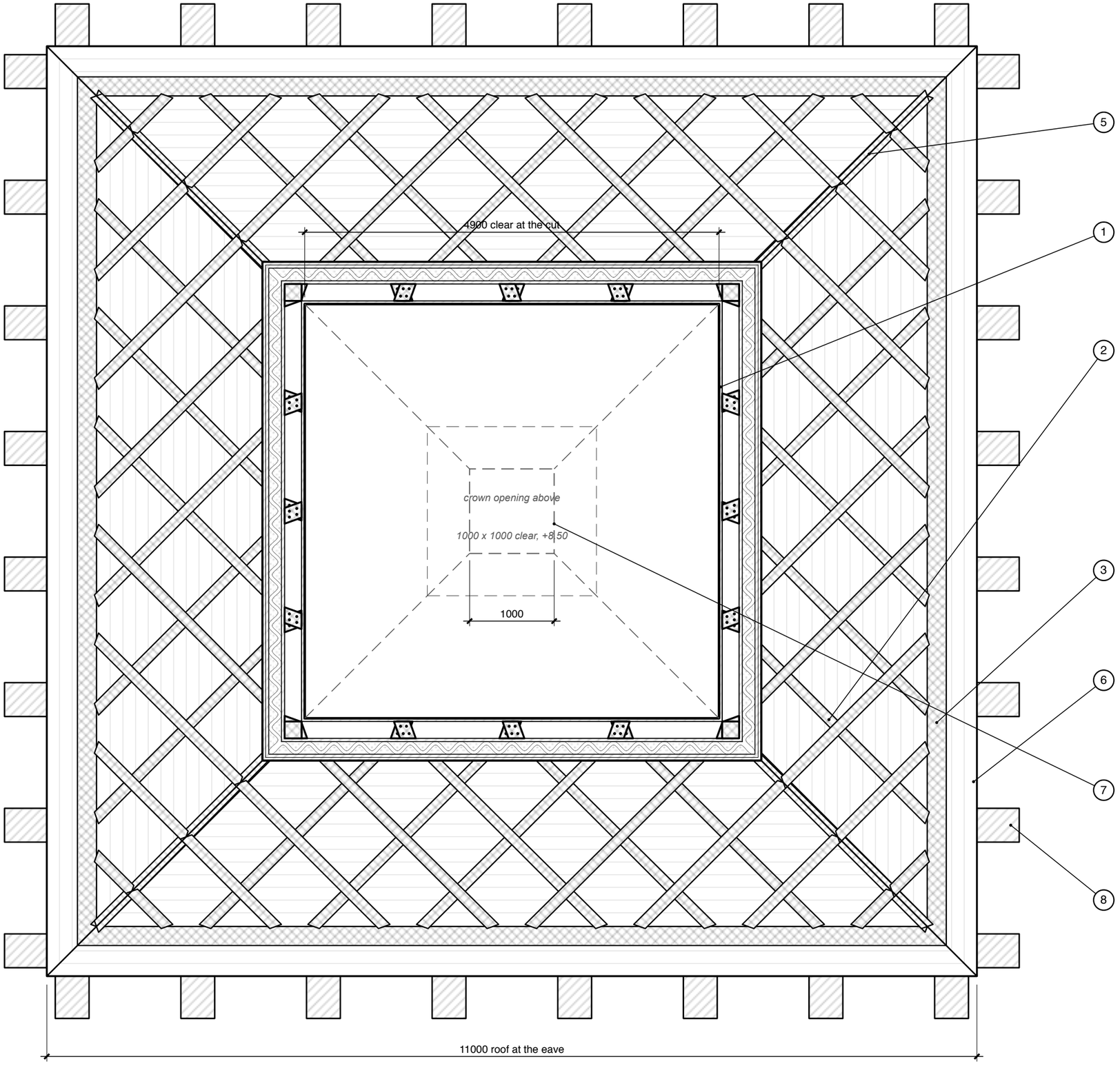
0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · dimensions in mm

2026-10-14 · hand estimates, not engineered

06



Roof plan, cut at +5.90 · grid B

horizontal cut through the roof, looking down on the ribs

1. Cut at +5.90 (3400 above the eave, just below the horizontal rib at +6.02), room side out: inner bead 30, rib zone 160 with the cast ribs cut (crossed hatch; empty cavity between them), middle bead 30, lattice zone 120 (wavy bead + insulation, dotted), 2 outer beads 30 (exposed concrete).
2. Ribs of grid B seen below the cut: the two families of diagonals from the base ring at the eave up to the cut and the hips.
3. Base ring: the 300 solid base at the eave (+2.50 to +2.80).
4. No horizontal ribs in grid B between the base ring and the top ring.
5. Hips: ribs from the corners of the eave up to the cut; dashed above it, up to the crown opening.
6. Roof edge at the eave (outer face of the outer beads). Very faint squares = printed layer joints (real layers 10, drawn every 200 in height).
7. Above the cut (removed, dashed): hips and the crown opening at +8.50, 1000 clear inside (1275 at the rib axis, 2000 outside the outer beads).
8. Buttresses below the roof: their ends show beyond the roof edge; the wall under the roof is dashed.

Notes

Matches the roof-grid elevations: same rib geometry, same levels (+4.26, +6.02, +7.68 for the horizontal ribs of grid A).

The ribs lie in the rib zone of the shell, between the inner and the middle bead; here they are drawn as if the middle bead, the lattice zone and the outer beads were transparent.

0 1 2 3 4 m

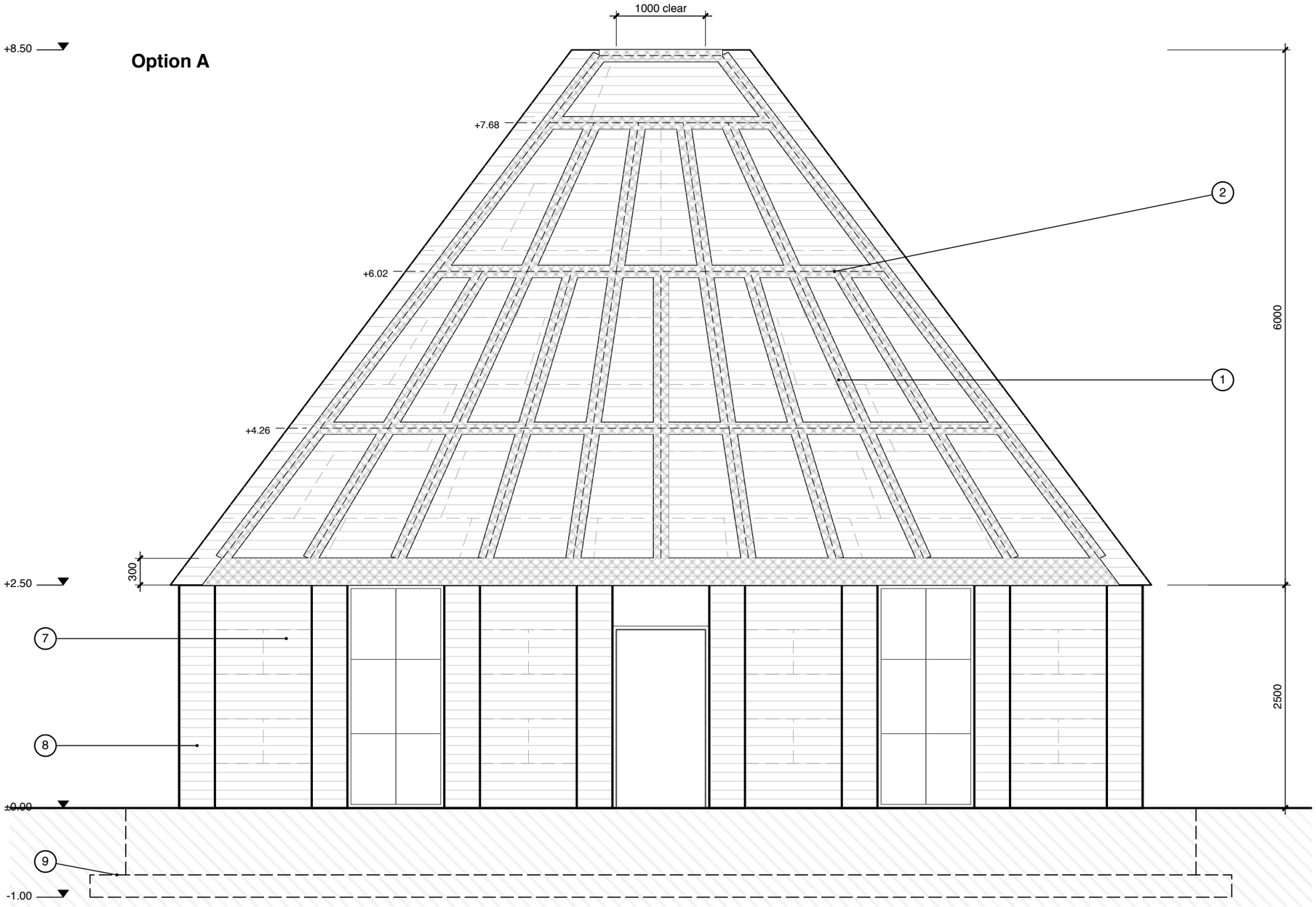
Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · dimensions in mm

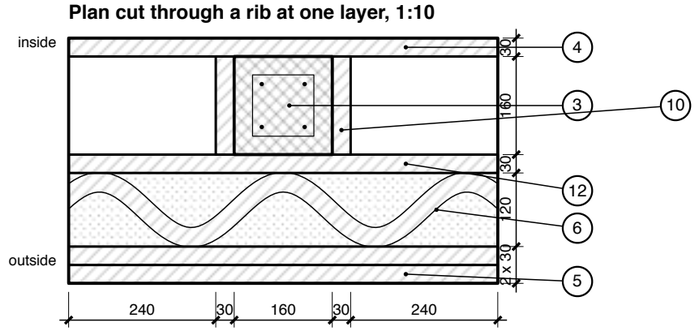
2026-10-14 · hand estimates, not engineered

07

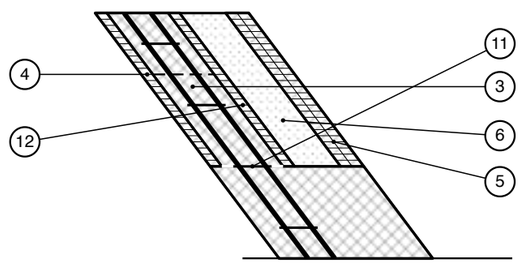
Option A



Plan cut through a rib at one layer, 1:10



Section along a rib at the eave, 1:20



Roof grid A: ruled-surface columns

one roof face seen square on

1. Columns run in straight lines from the base edge (every 1028) towards the crown edge, so they converge like the ruled surface of the sketch.
2. When the spacing drops to 500 (about +6.02) every other column stops; at about +7.68 only the four hip columns go on to the crown ring.
3. Rib (column) 160 wide x 160 deep in the rib zone, between the inner and the middle bead: the printed channel stays as formwork, filled with cast concrete; 4 Ø16 + Ø8 stirrups every 200 (assumed), or fibre concrete (see notes).
4. Inner bead 30: continuous, covers the grid on the room side.
5. 2 outer beads 30: continuous, the outside of the printed shell, exposed concrete (no insulation or cladding outside).
6. Lattice zone 120: wavy printed bead 30 touching the middle and the outer beads in turn, insulation between (dotted).
7. Printed wall 400, to +2.50: inner bead 30, lattice zone 310 (wavy bead + insulation), 2 outer beads 30.
8. Buttresses 400 wide, 1000 from the inner face (600 outside), about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included); every other bay is printed wall, the bays between are full-height curtain wall (door in the middle one).
9. Strip footing 1800 x 250, underside at -1.00 (hidden, dashed): printed edge formwork, cast core.
10. Printed channel walls 30 each side of the rib, from the inner to the middle bead: permanent formwork. Between the ribs the rib zone is an empty cavity.
11. Poured core: the solid base (300, full concrete over the whole 400) and the rib in one pour, lifts of about 600 from the eave (assumed, dashed), no joint between base and rib. Printed layer joints every 10 (drawn every 20).
12. Middle bead 30: closes the rib zone outside; inner and middle bead are the structural skins, the ribs the webs (as in the Karamba model).

Notes

Columns stop on a horizontal rib (at 3517 and 5183 above the eave); bars are lapped / hooked into it.

Horizontal ribs at 1758, 3517, 5183, plus the 300 solid base (bottom ring) and the top ring at the crown.

Dashed lines = printed pieces in running bond (about 1500 x 750); very faint lines = printed layer joints (10 mm layers, drawn every 100).

Printed mortar beads are permanent formwork; ribs, base and buttress cores are poured concrete (after the Obayashi 3dpod paper, 2023). Steel-fibre concrete could replace the rib bars, to be confirmed by the engineer.

Rough quantities, four faces: ribs 318 m, steel 2005 kg (4 Ø16 per rib), poured concrete 8.1 m3.

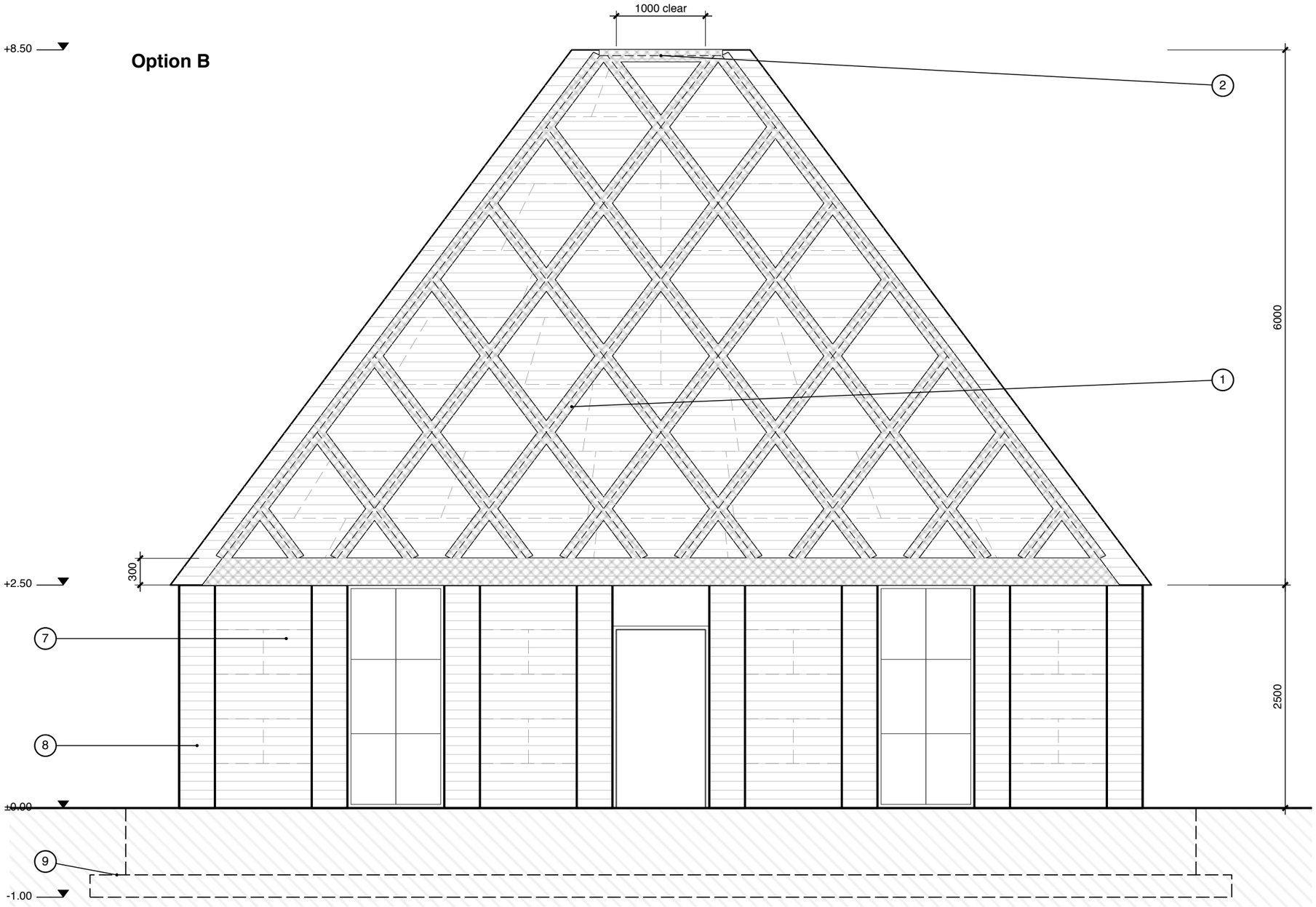
0 1 2 3 4 m

Pyramid roof · 3D-printed concrete · GSAPP

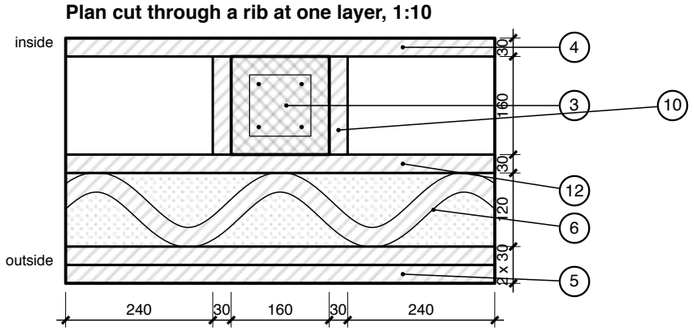
Scale 1:50 at A3 · rib plan cut 1:10, rib section 1:20 · dimensions in mm

2026-10-14 · hand estimates, not engineered

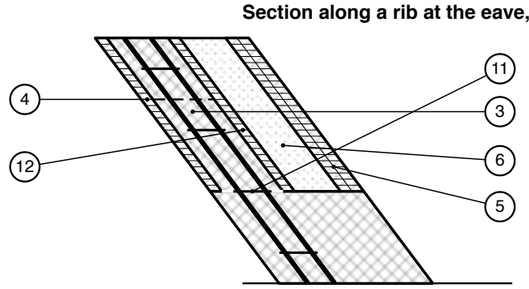
Option B



Plan cut through a rib at one layer, 1:10



Section along a rib at the eave, 1:20



Roof grid B: diamond lattice

one roof face seen square on

1. Diagonals parallel to the two hips (same inclination as the roof edges, 59.0° in the face), every 1284 along the base: equal diamonds 1284 wide.
2. Bottom ring = the 300 solid base; top ring at the crown. The hips are two of the diagonals. No other horizontal ribs.
3. Rib (column) 160 wide x 160 deep in the rib zone, between the inner and the middle bead: the printed channel stays as formwork, filled with cast concrete; 4 Ø16 + Ø8 stirrups every 200 (assumed), or fibre concrete (see notes).
4. Inner bead 30: continuous, covers the grid on the room side.
5. 2 outer beads 30: continuous, the outside of the printed shell, exposed concrete (no insulation or cladding outside).
6. Lattice zone 120: wavy printed bead 30 touching the middle and the outer beads in turn, insulation between (dotted).
7. Printed wall 400, to +2.50: inner bead 30, lattice zone 310 (wavy bead + insulation), 2 outer beads 30.
8. Buttresses 400 wide, 1000 from the inner face (600 outside), about every 1.5 m (7 bays of 1486 on the wall centreline, 8 per side, corners included); every other bay is printed wall, the bays between are full-height curtain wall (door in the middle one).
9. Strip footing 1800 x 250, underside at -1.00 (hidden, dashed): printed edge formwork, cast core.
10. Printed channel walls 30 each side of the rib, from the inner to the middle bead: permanent formwork. Between the ribs the rib zone is an empty cavity.
11. Poured core: the solid base (300, full concrete over the whole 400) and the rib in one pour, lifts of about 600 from the eave (assumed, dashed), no joint between base and rib. Printed layer joints every 10 (drawn every 20).
12. Middle bead 30: closes the rib zone outside; inner and middle bead are the structural skins, the ribs the webs (as in the Karamba model).

Notes

Every diagonal ends on a ring or on a hip, so no bar has a free end in the shell.

Diagonals do not stop as the roof narrows: the diamonds just get clipped by the hips.

Dashed lines = printed pieces in running bond (about 1500 x 750); very faint lines = printed layer joints (10 mm layers, drawn every 100).

Printed mortar beads are permanent formwork; ribs, base and buttress cores are poured concrete (after the Obayashi 3dpod paper, 2023). Steel-fibre concrete could replace the rib bars, to be confirmed by the engineer.

Rough quantities, four faces: ribs 326 m, steel 2055 kg (4 Ø16 per rib), poured concrete 8.3 m3.

0 1 2 3 4 m

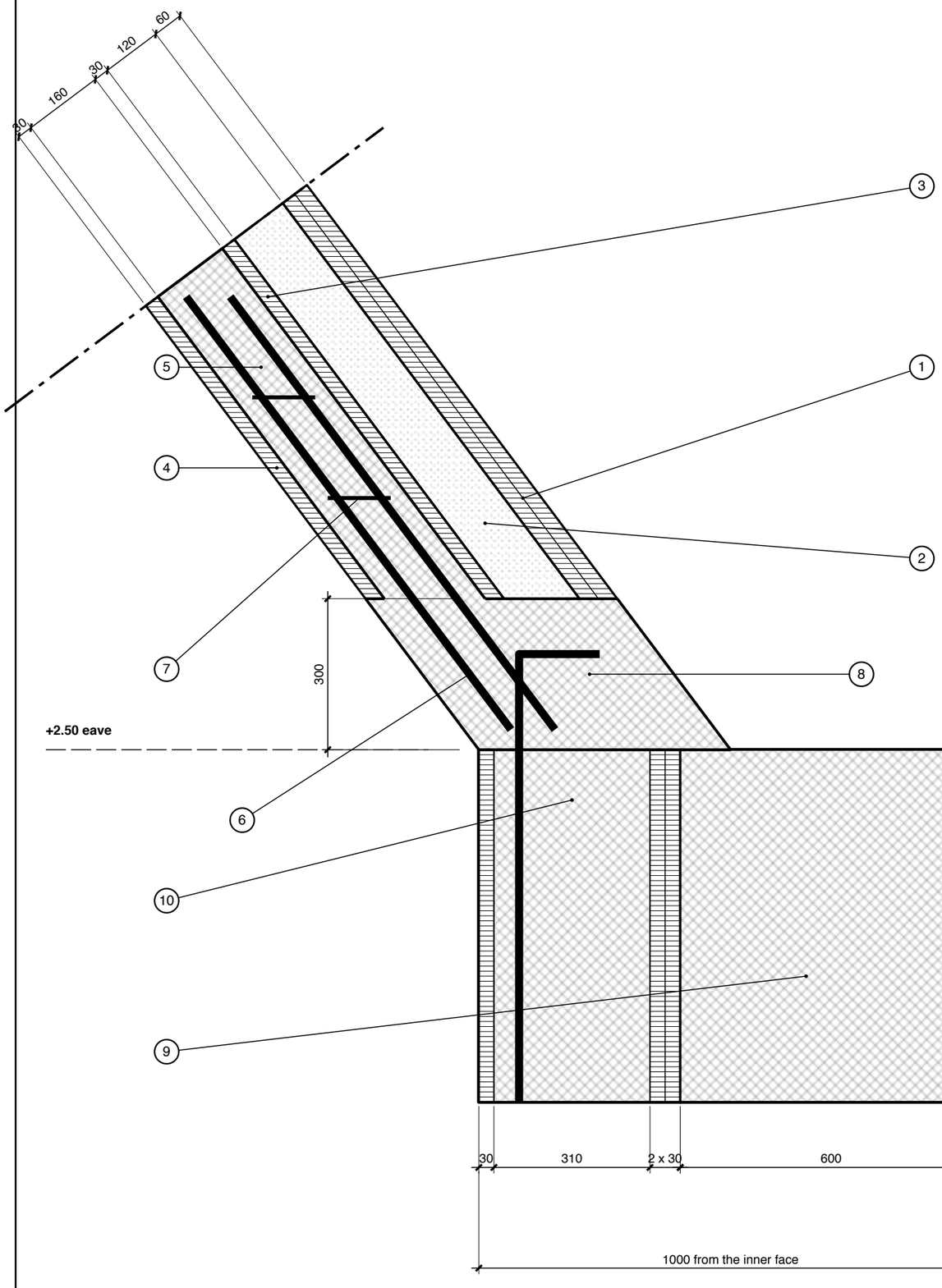
Pyramid roof · 3D-printed concrete · GSAPP

Scale 1:50 at A3 · rib plan cut 1:10, rib section 1:20 · dimensions in mm

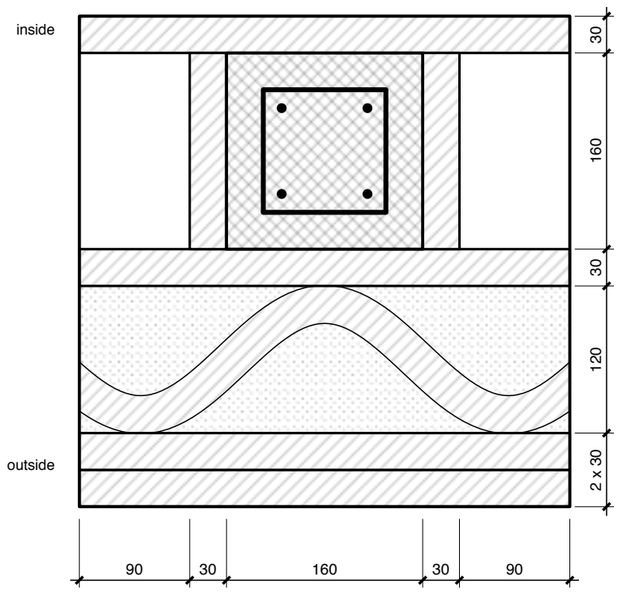
2026-10-14 · hand estimates, not engineered

09

Detail D1: eave, roof to wall



Plan cut through a rib at one layer, 1:5



1. 2 outer beads 30, the outside of the printed piece: exposed concrete, no insulation or cladding outside (the insulation is in the lattice zone).
2. Lattice zone 120: a wavy printed bead 30 that touches the middle bead and the outer beads in turn (period about 300, assumed), insulation between its loops (dotted).
3. Middle bead 30: closes the rib zone on the outside. Inner and middle bead are the two structural skins of the roof, the ribs the webs between them (as in the Karamba model).
4. Inner bead 30, room side: continuous, it covers the ribs. The roof is 400 thick square to the slope from eave to crown.
5. Rib (cast column) 160 wide x 160 deep: fills the rib zone between the inner and the middle bead, inside two printed channel walls of 30 (see the plan cut), filled with poured concrete; section cut along the rib. It rises out of the solid base in the same pour, without a joint. Between the ribs the rib zone is an empty cavity.
6. Rib bars R, 4 Ø16 (2 in the cut), anchored in the solid base where they cross the hooks of the wall bars A (assumed; steel fibre in the concrete could replace them).
7. Stirrups Ø8 every 200 (assumed).
8. Solid base of the roof: the first 300 (vertical) above the eave is full cast concrete over the whole 400, no cavity or lattice, poured in one go with the rib above it; wall bars A hook into it. Replaces the ring beam.
9. Buttress 400 wide, 1000 from the inner face (600 beyond the wall): printed formwork (60 end cap of 2 beads shown) filled with cast concrete; top flush with the roof base at +2.50.
10. Wall 400 at the buttress: printed inner bead 30 and 2 outer beads 30, the 310 zone between them cast and one body with the buttress (no insulation here, the cut is through the buttress; between buttresses this zone is the wavy bead + insulation). Wall bars A Ø16 run in the cast zone from the footing up into the solid base and end in a 160 hook (assumed). The roof grid is A or B (see the roof-grid sheets); the detail is cut along a rib.

0 1 2 3 4 dm

Pyramid roof · 3D-printed concrete · GSAPP
Scale 1:10 at A3 · rib plan cut 1:5 · dimensions in mm
2026-10-14 · hand estimates, not engineered